17 When an object of mass m is raised through a vertical height Δh, the gain of its gravitational potential energy is ΔE_P.

ΔE_P and Δh are related by the equation

$$\Delta E_P = mg\Delta h,$$

where g is the acceleration of free fall.

The definition of which physical quantity is needed to derive this equation?

A acceleration

B momentum

C power

D work done